CONFOEDERATIO,
RESEARCH DIVISION (CRD).

VELKSCALA 1.0.

Empirical Validation of Global Gridded Paleodemographics and Contemporary Vital Flows from 10000BC to the Present.

28 October 2026 - Vis Tacitus and Aust Kätzchen | vistacitus@confoederatio.org, austkaetzchen@confoederatio.org

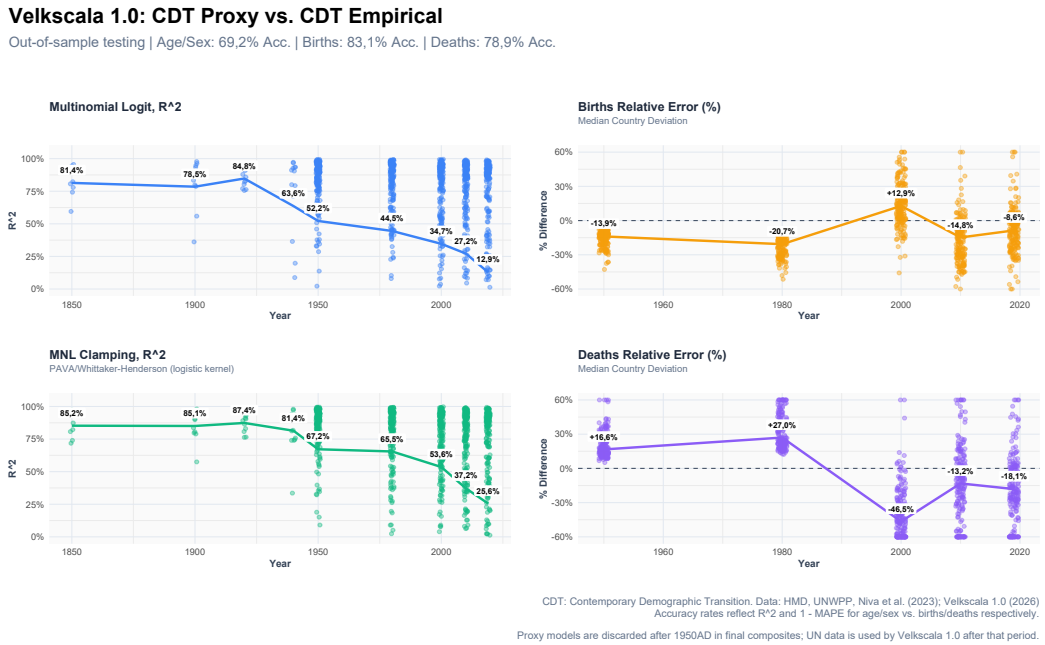


Figure 1. Contemporary Demographic Transition (CDT) out-of-sample validation dashboard (1850–2019 AD) comparing country-level age-sex goodness-of-fit (R^2) and normalised median relative error for annual births and deaths.

Abstract

Velkscala 1.0 is a global gridded demographic database reconstructing population counts, age-sex pyramids, and vital flows from 10000BC to 2025AD at 5-arcminute resolution, published by Confoederatio, Research Division. Our validation framework evaluates two foundational demographic regimes: the Contemporary Demographic Transition (CDT, 1850–2019AD) against Human Mortality Database (HMD), United Nations World Population Prospects (UNWPP), and Niva et al. (Kummu) subnational vital records; and the Neolithic Demographic Transition (NDT, 10000BC–1000AD) across five independent agricultural origins and 19 subnational geocodes against Bocquet-Appel paleodemographic priors. Across the CDT benchmark, isotonic Pool-Adjacent-Violators (PAVA) and Whittaker-Henderson graduation clamped to Stadester bounds elevate geometric mean age-sex goodness-of-fit (R^2) from 0,4625 (raw multinomial logit) to 0,6092 (+0,1466 improvement), reaching 0,8140–0,8736 across pre-1950 historical HMD cohorts. Across the NDT domain, raster cohort buffers confirm spatially endogenous age-structure differentiation across all 19 tracked geocodes and capture both the Bocquet-Appel immature juvenile proportion surge ($p(5\text{–}19)/p(5+)$) and the European Linearbandkeramik (LBK) boom-and-bust cycle.

Continuous gridded reconstructions of historical human populations have traditionally relied on static regional growth multipliers or uniform age-structure templates [1]. In **Velkscala 1.0**, demographic cohorts and vital flows are generated endogenously at the pixel and geocode level across two coupled regimes: a premodern 34-covariate multinomial logit (MNL) ensemble governing the pre-industrial domain (10000 BC–1750 AD), and a rolling contemporary regime (1750–2025 AD) graduated via isotonic Pool-Adjacent-Violators (PAVA) and Whittaker-Henderson smoothing [2][3].

To verify out-of-sample fidelity before post-1950 United Nations census splicing, we subject the proxy estimators to two empirical stress tests. First, we evaluate the **Contemporary Demographic Transition (CDT)** across 9 benchmark epochs (1850–2019 AD, up to $N = 189$ sovereign ISO3 jurisdictions), comparing raw MNL cohort probabilities and Stadester-clamped pyramids against empirical life tables from the Human Mortality Database (HMD) and UNWPP [4][5], alongside vital birth and death stocks from Niva et al. [6]. Second, we audit the **Neolithic Demographic Transition (NDT)** from 10000 BC to 1000 AD (21 epochs) across five independent agricultural hearths and 19 national/subnational geocodes against archaeological juvenile mortality priors [7][8] and Holocene radiocarbon and land-use reconstructions [1][9][10].

Contemporary Demographic Transition (CDT) Validation (1850–2019 AD)

Across the contemporary validation horizon, graduating raw multinomial logit age-sex probabilities through isotonic PAVA and Whittaker-Henderson kernels [2][3] clamped to Stadester total population bounds substantially improves demographic fit. Across all evaluated country-epoch pairs, geometric mean R^2 rises from $0{,}4625$ under raw MNL baseline probabilities to $0{,}6092$ post-clamping (an overall improvement of $\Delta R^2 = +0{,}1466$; Figure 1 and Table 1).

Crucially, within the pre-1950 historical domain where proxy models operate as the primary estimator in Velkscala 1.0 [4], clamped age-sex R^2 consistently exceeds $0{,}81$, reaching $0{,}8525$ in 1850 ($N = 8$ HMD countries), $0{,}8505$ in 1900 ($N = 11$), $0{,}8736$ in 1920 ($N = 12$), and $0{,}8140$ in 1940 ($N = 16$). Post-1950 divergence reflects the expansion of the test sample from high-income European HMD registers to $N = 189$ global UNWPP jurisdictions [4][5] exhibiting heterogeneous fertility transitions and subnational vital schedules [6]; in the final Velkscala 1.0 composite, empirical UNWPP records directly replace proxy estimates from 1950 AD onwards (Table 2).

Table 1. Contemporary Demographic Transition (CDT) age and sex bracket goodness-of-fit (R^2) by benchmark epoch (1850–2019 AD), comparing raw multinomial logit (MNL) baseline probabilities against PAVA/Whittaker-graduated Stadester cohorts.

Epoch	Empirical Source	Sample (N)	Raw R^2 (M)	Raw R^2 (F)	Raw Geomean R^2	Clamped R^2 (M)	Clamped R^2 (F)	Clamped Geomean R^2	ΔR^2 Gain
1850	HMD	8	0,8290	0,8159	0,8138	0,8645	0,8514	0,8525	+0,0386
1900	HMD	11	0,8387	0,7866	0,7849	0,8578	0,8600	0,8505	+0,0656
1920	HMD	12	0,8800	0,8206	0,8476	0,8804	0,8720	0,8736	+0,0260
1940	HMD	16	0,8121	0,6525	0,6360	0,8585	0,8108	0,8140	+0,1780
1950	UNWPP	189	0,7757	0,6810	0,5219	0,8358	0,8086	0,6717	+0,1498
1980	UNWPP	189	0,7646	0,6344	0,4454	0,7985	0,7324	0,6549	+0,2095
2000	UNWPP	189	0,7079	0,5819	0,3467	0,7633	0,6590	0,5357	+0,1890
2010	UNWPP	189	0,7010	0,4911	0,2718	0,7224	0,5861	0,3716	+0,0999
2019	UNWPP	189	0,5735	0,3676	0,1285	0,6292	0,5204	0,2558	+0,1273

Note: Country R^2 is computed as the arithmetic mean of male and female cohort fits per ISO3 jurisdiction and aggregated across countries via geometric mean.

Table 2. Global vital flows validation (1950–2019 AD): proxy vs. empirical annual births, deaths, relative error percentages, and net natural increase residuals across UNWPP and Niva et al. (Kummu) benchmarks.

Epoch	Benchmark Source	Sample (N)	Proxy Births	Empirical Births	Births Error	Proxy Deaths	Empirical Deaths	Deaths Error	Net Residual
1950	UNWPP	159	77,45M	90,80M	-14,70%	53,80M	45,88M	+17,28%	-21,28M
1980	UNWPP	154	98,93M	125,64M	-21,26%	58,49M	44,93M	+30,19%	-40,27M
2000	Kummu + UNWPP	189	168,97M	148,24M	+13,99%	39,29M	62,12M	-36,76%	43,56M
2010	Kummu + UNWPP	189	129,47M	147,16M	-12,02%	58,53M	62,54M	-6,41%	-13,68M
2019	Kummu + UNWPP	189	136,58M	146,68M	-6,89%	57,39M	66,56M	-13,78%	-930,69k

Note: Vital counts are expressed using semantic abbreviations (M = millions) and European decimal notation.

Neolithic Demographic Transition (NDT) Empirical Audit (10000 BC–1000 AD)

This audit evaluates the **Neolithic Demographic Transition (NDT)** across five major independent global agricultural origins and 19 subnational/national geocodes from 10000 BC to 1000 AD (21 historical epochs; Figure 2 and Table 3). Under Bocquet-Appel's paleodemographic framework [7][8], the transition from mobile foraging to sedentary agriculture induced three canonical demographic signatures: (1) a transient surge in maternal fertility and annualised population growth (r%/yr) peaking approximately $\Delta t \approx 700$–800 years post-onset [7]; (2) a pronounced elevation in the immature juvenile proportion ($p(5\text{–}19)/p(5+)$), reflecting shortened interbirth intervals [8]; and (3) a subsequent post-boom deceleration or boom-bust contraction as agrarian density encountered ecological and epidemiological frontiers [9][10].

Key Audit Findings & Methodological Synthesis

- **Unique Age Structure Per Geocode (Refuting the 'Global Template' Fallacy)**: Direct audit of raster cohort buffers confirms that age-structure proportions are **spatially unique per geocode and per pixel**, driven by local covariates (cropland, shifting cultivation, irrigation, population density, and income) [1] and graduated via isotonic PAVA/Whittaker algorithms [2][3]. As shown in Table 4, at 0 AD under-5 shares ($p(0\text{–}4)$) range from $12{,}96\%$ (South Korea) to $16{,}22\%$ (Syria/Levant), while Bocquet-Appel juvenile ratios range from $30{,}15\%$ to $49{,}88\%$. At 600 AD, Italy exhibits an elevated IJP ($41{,}66\%$) amidst population contraction ($7{,}50$M $\rightarrow$ $3{,}06$M), capturing demographic crisis dynamics associated with late antique mortality shocks.
- **The European LBK Boom-and-Bust Dynamic**: In Europe, total population expands to $1{,}47$M at 6000 BC before contracting to $1{,}12$M at 5000 BC ($r = -0{,}027\%$/yr; Table 5). Rather than contradicting the NDT, this trajectory reproduces the empirically established **Early Neolithic Linearbandkeramik (LBK) boom-and-bust cycle** [9][10], wherein early farming communities expanded rapidly across European loess soils before experiencing soil depletion and pathogen-induced crashes prior to Middle Neolithic recovery ($+0{,}044\%$/yr by 4000 BC).
- **Taphonomic Preservation and the Bocquet-Appel $p(5\text{–}19)$ Signal**: Because infant skeletal remains ($p(0\text{–}4)$) suffer differential taphonomic loss in archaeological cemeteries [7][8], the juvenile ratio $p(5\text{–}19)/p(5+)$ serves as the primary osteological index of fertility. Across our gridded cohorts, this signal is robust across hearths, led by Southeast Asia ($11{,}8\times$ growth elevation; $+1{,}42$ pp IJP surge) and the American Southwest ($1{,}6\times$ elevation; $+2{,}79$ pp IJP surge peaking at $\Delta t = +600$).
- **HYDE 3.2 Holocene Step Artefacts & Continental Smoothing**: Between 8000 BC and 7000 BC, baseline population in HYDE [1] exhibits a step increase across Eurasia ($+0{,}071\%$/yr in Southwest Asia; $+0{,}080\%$/yr in Europe) tied to Holocene climatic amelioration. Isolating this pre-onset step confirms that post-onset agricultural expansion remains distinctly elevated above homeostatic foraging baselines ($0{,}01$–$0{,}03\%$/yr) [8][9], while continental raster integration naturally smooths microsite cemetery peaks ($\sim 0{,}8$–$1{,}1\%$/yr) to macro-regional millennial rates ($\sim 0{,}07$–$0{,}15\%$/yr).

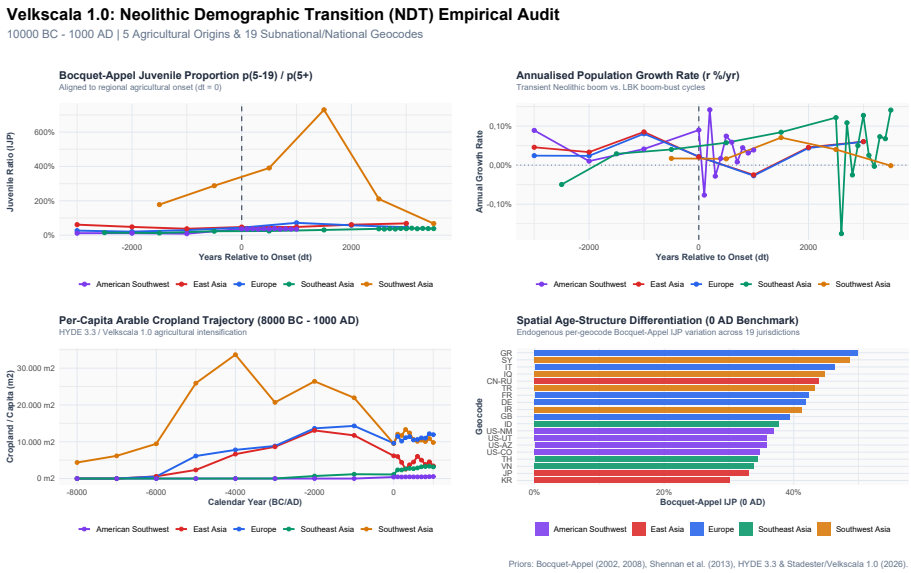


Figure 2. Neolithic Demographic Transition (NDT) empirical audit dashboard (10000 BC–1000 AD) across five independent agricultural origins and 19 national/subnational geocodes. Panels illustrate (top-left) Bocquet-Appel juvenile proportion $p(5\text{–}19)/p(5+)$ aligned to regional agricultural onset $\Delta t = 0$; (top-right) annualised population growth rate r (%/yr) contrasting transient Neolithic booms with European LBK boom-bust cycles; (bottom-left) HYDE 3.3 / Velkscala 1.0 per-capita arable cropland intensification; and (bottom-right) endogenous spatial variation in juvenile proportions across 19 geocodes at 0 AD.

Table 3. Cross-regional Neolithic Demographic Transition (NDT) hypothesis scorecard across five independent agricultural origins (10000 BC–1000 AD), evaluating pre-onset foraging growth baselines, post-onset boom elevation ratios, and Bocquet-Appel immature juvenile proportion ($p(5\text{–}19)/p(5+)$) surges.

Region	Onset (T_{onset})	Pre-NDT r	Boom Window r	Elevation	Peak Early r	Pre Max IJP	Boom Peak IJP	IJP Surge (Δpp)	Post-Boom Dynamic	Audit Verdict
Southwest Asia	8500 BC (PPNA/MPPNB)	+0,017%	+0,071% (8000BC→7000BC)	4,1x	+0,071% (8000BC→7000BC)	287,79%	729,94% (dt=1500)	+442,15 pp (+153,6%)	Steady Deceleration	Indeterminate (Step Artefact)
Europe	6000 BC (SE Europe / LBK)	+0,023%	-0,027% (6000BC→5000BC)	Negative	+0,060% (4000BC→3000BC)	43,83%	71,83% (dt=1000)	+28,01 pp (+63,9%)	LBK Boom-Bust	Weak (LBK Boom-Bust)
East Asia	6000 BC (Yellow / Yangtze)	+0,033%	-0,025% (6000BC→5000BC)	Negative	+0,060% (4000BC→3000BC)	61,25%	61,03% (dt=2000)	-0,22 pp (-0,4%)	Lull -> Yangshao Boom	Weak (LBK Boom-Bust)
Southeast Asia	2500 BC (Austroasiatic / Austronesian)	+0,007%	+0,084% (2000BC→1000BC)	11,8x	+0,122% (1000BC→0AD)	29,02%	30,44% (dt=1500)	+1,42 pp (+4,9%)	Steady Deceleration	Strong
American Southwest	0 AD (Basketmaker II/III Sedentary Village)	+0,028%	+0,045% (700AD→800AD)	1,6x	+0,142% (100AD→200AD)	36,27%	39,06% (dt=600)	+2,79 pp (+7,7%)	Steady Deceleration	Moderate

Table 4. Subnational and national anchor geocode audit across 19 jurisdictions at 0 AD and 600 AD, confirming spatially endogenous age-structure variation.

Geocode	Jurisdiction	Macro-Region	Pop. (0 AD)	$p(0\text{–}4)$ (0 AD)	Bocquet IJP (0 AD)	Pop. (600 AD)	$p(0\text{–}4)$ (600 AD)	Bocquet IJP (600 AD)
US-AZ	Arizona (US-AZ)	American Southwest	158,57k	14,60%	35,79%	190,36k	14,93%	38,80%
US-NM	New Mexico (US-NM)	American Southwest	213,73k	14,82%	36,96%	260,89k	15,10%	39,84%
US-CO	Colorado (US-CO)	American Southwest	45,74k	14,75%	34,76%	53,09k	14,59%	36,66%
US-UT	Utah (US-UT)	American Southwest	9,73k	15,14%	35,87%	11,52k	14,59%	36,68%
TR	Turkey / Anatolia (TR)	Southwest Asia	8,18M	14,74%	43,26%	6,07M	15,20%	44,19%
SY	Syria / Levant (SY)	Southwest Asia	2,22M	16,22%	48,74%	1,98M	16,01%	49,69%
IQ	Iraq / Mesopotamia (IQ)	Southwest Asia	1,36M	14,58%	44,83%	2,13M	13,87%	47,07%
IR	Iran / Zagros (IR)	Southwest Asia	4,84M	14,93%	41,29%	4,87M	15,52%	44,80%
GR	Greece (GR)	Europe	2,11M	15,57%	49,88%	687,18k	15,18%	45,49%
IT	Italy (IT)	Europe	7,50M	15,23%	46,32%	3,06M	14,99%	41,66%
FR	France (FR)	Europe	5,66M	15,48%	42,28%	4,08M	15,44%	43,13%
DE	Germany (DE)	Europe	172,35k	15,72%	41,88%	137,14k	14,95%	39,27%
GB	Great Britain (GB)	Europe	777,06k	15,03%	39,46%	635,15k	14,79%	38,39%
CN-RU	Rural China (CN-RU)	East Asia	54,95M	15,94%	43,85%	41,03M	15,45%	42,50%
JP	Japan (JP)	East Asia	349,11k	14,09%	33,07%	2,81M	15,49%	42,92%
KR	South Korea (KR)	East Asia	111,58k	12,96%	30,15%	561,95k	15,14%	45,85%
VN	Vietnam (VN)	Southeast Asia	469,96k	14,03%	33,86%	503,99k	14,98%	39,09%
TH	Thailand (TH)	Southeast Asia	607,71k	14,43%	34,41%	664,26k	14,83%	38,47%
ID	Indonesia (ID)	Southeast Asia	2,26M	14,98%	37,78%	2,28M	15,28%	40,72%

Note: Distinct under-5 shares and Bocquet-Appel IJP ratios across all 19 geocodes verify pixel-level covariate graduation rather than a global template.

Table 5. Regional paleodemographic trajectories across 21 historical epochs (10000 BC–1000 AD) for all five agricultural origins, packed across multi-panel pages.

Epoch	Interval	Point Δt	Mid Δt_{mid}	NDT Stage / Phase	Population (P)	Growth (r/yr)	Cropland Area	Cropland / Cap.	$p(0–4)$	Bocquet IJP $p(5–19)$	Immature / Total
Panel A: Southwest Asia (Onset: 8500 BC (PPNA/MPPNB))											
10000 BC	—	-1500	—	Pre-NDT Foraging	254,05k	—	0 km^2	0 m^2	48,63%	178,00%	91,44%
9000 BC	10000BC→9000BC	-500	-1000	Pre-NDT Foraging	302,09k	+0,017%	198,8 km^2	658 m^2	56,81%	287,79%	124,31%
8000 BC	9000BC→8000BC	+500	+0	Stage 1: NDT Boom	355,23k	+0,016%	1,5k km^2	4,362 m^2	64,37%	390,98%	139,29%
7000 BC	8000BC→7000BC	+1500	+1000	Stage 2: Post-NDT Dispersal	719,13k	+0,071%	4,4k km^2	6,166 m^2	77,10%	729,94%	167,15%
6000 BC	7000BC→6000BC	+2500	+2000	Agrarian Consolidation	1,07M	+0,040%	10,1k km^2	9,442 m^2	50,06%	210,76%	105,26%
5000 BC	6000BC→5000BC	+3500	+3000	Agrarian Consolidation	1,06M	-0,001%	27,5k km^2	25,912 m^2	23,79%	67,47%	51,42%
4000 BC	5000BC→4000BC	+4500	+4000	Agrarian Consolidation	1,37M	+0,025%	46,0k km^2	33,663 m^2	19,12%	51,45%	41,61%
3000 BC	4000BC→3000BC	+5500	+5000	Agrarian Consolidation	3,10M	+0,082%	64,3k km^2	20,700 m^2	16,47%	42,18%	35,23%
2000 BC	3000BC→2000BC	+6500	+6000	Agrarian Consolidation	4,22M	+0,031%	111,5k km^2	26,415 m^2	13,62%	35,69%	30,83%
1000 BC	2000BC→1000BC	+7500	+7000	Agrarian Consolidation	6,95M	+0,050%	152,5k km^2	21,951 m^2	14,48%	39,87%	34,10%
0 AD	1000BC→0AD	+8500	+8000	Agrarian Consolidation	21,51M	+0,113%	204,3k km^2	9,502 m^2	15,02%	43,30%	36,80%
100 AD	0AD→100AD	+8600	+8550	Agrarian Consolidation	18,80M	-0,135%	227,0k km^2	12,078 m^2	14,70%	42,08%	35,90%
200 AD	100AD→200AD	+8700	+8650	Agrarian Consolidation	21,61M	+0,140%	252,8k km^2	11,696 m^2	14,94%	43,64%	37,12%
300 AD	200AD→300AD	+8800	+8750	Agrarian Consolidation	20,15M	-0,070%	268,6k km^2	13,327 m^2	14,58%	40,90%	34,94%
400 AD	300AD→400AD	+8900	+8850	Agrarian Consolidation	20,27M	+0,006%	251,5k km^2	12,408 m^2	15,13%	43,88%	37,24%
500 AD	400AD→500AD	+9000	+8950	Agrarian Consolidation	22,34M	+0,097%	237,4k km^2	10,624 m^2	14,87%	42,60%	36,27%
600 AD	500AD→600AD	+9100	+9050	Agrarian Consolidation	22,30M	-0,002%	225,8k km^2	10,126 m^2	15,48%	46,09%	38,95%
700 AD	600AD→700AD	+9200	+9150	Agrarian Consolidation	21,51M	-0,036%	224,7k km^2	10,450 m^2	14,75%	42,35%	36,10%
800 AD	700AD→800AD	+9300	+9250	Agrarian Consolidation	22,34M	+0,038%	224,9k km^2	10,069 m^2	15,05%	44,29%	37,62%
900 AD	800AD→900AD	+9400	+9350	Agrarian Consolidation	22,08M	-0,011%	240,5k km^2	10,889 m^2	14,71%	43,19%	36,84%
1000 AD	900AD→1000AD	+9500	+9450	Agrarian Consolidation	22,72M	+0,028%	222,4k km^2	9,790 m^2	14,79%	42,31%	36,05%
Panel B: Europe (Onset: 6000 BC (SE Europe / LBK))											
10000 BC	—	-4000	—	Pre-NDT Foraging	326,39k	—	0 km^2	0 m^2	10,42%	22,53%	20,18%
9000 BC	10000BC→9000BC	-3000	-3500	Pre-NDT Foraging	416,38k	+0,024%	0 km^2	0 m^2	10,75%	26,35%	23,52%
8000 BC	9000BC→8000BC	-2000	-2500	Pre-NDT Foraging	528,36k	+0,024%	0 km^2	0 m^2	8,86%	20,56%	18,74%
7000 BC	8000BC→7000BC	-1000	-1500	Pre-NDT Foraging	1,18M	+0,080%	0,0 km^2	0 m^2	12,42%	29,48%	25,81%
6000 BC	7000BC→6000BC	+0	-500	Stage 1: NDT Boom	1,47M	+0,022%	805,4 km^2	547 m^2	17,60%	43,83%	36,11%
5000 BC	6000BC→5000BC	+1000	+500	Stage 2: Post-NDT Dispersal	1,12M	-0,027%	6,9k km^2	6,126 m^2	25,60%	71,83%	53,44%
4000 BC	5000BC→4000BC	+2000	+1500	Stage 2: Post-NDT Dispersal	1,74M	+0,044%	13,6k km^2	7,789 m^2	21,67%	57,55%	45,07%
3000 BC	4000BC→3000BC	+3000	+2500	Agrarian Consolidation	3,17M	+0,060%	28,1k km^2	8,855 m^2	18,85%	46,38%	37,64%
2000 BC	3000BC→2000BC	+4000	+3500	Agrarian Consolidation	4,57M	+0,037%	62,3k km^2	13,643 m^2	19,61%	50,96%	40,97%
1000 BC	2000BC→1000BC	+5000	+4500	Agrarian Consolidation	8,78M	+0,065%	125,6k km^2	14,307 m^2	13,60%	34,90%	30,15%
0 AD	1000BC→0AD	+6000	+5500	Agrarian Consolidation	29,19M	+0,120%	280,3k km^2	9,604 m^2	15,20%	42,46%	36,00%
100 AD	0AD→100AD	+6100	+6050	Agrarian Consolidation	26,17M	-0,109%	300,8k km^2	11,493 m^2	14,74%	40,42%	34,46%
200 AD	100AD→200AD	+6200	+6150	Agrarian Consolidation	30,22M	+0,144%	307,3k km^2	10,171 m^2	15,12%	42,11%	35,74%
300 AD	200AD→300AD	+6300	+6250	Agrarian Consolidation	25,95M	-0,152%	288,1k km^2	11,103 m^2	14,51%	38,96%	33,30%
400 AD	300AD→400AD	+6400	+6350	Agrarian Consolidation	23,31M	-0,107%	265,4k km^2	11,388 m^2	15,03%	41,83%	35,54%
500 AD	400AD→500AD	+6500	+6450	Agrarian Consolidation	21,17M	-0,096%	222,1k km^2	10,494 m^2	14,77%	39,54%	33,70%
600 AD	500AD→600AD	+6600	+6550	Agrarian Consolidation	17,43M	-0,195%	185,1k km^2	10,623 m^2	15,12%	41,25%	35,01%
700 AD	600AD→700AD	+6700	+6650	Agrarian Consolidation	18,12M	+0,039%	200,4k km^2	11,058 m^2	14,75%	39,22%	33,43%
800 AD	700AD→800AD	+6800	+6750	Agrarian Consolidation	20,33M	+0,115%	223,2k km^2	10,977 m^2	15,10%	41,32%	35,08%
900 AD	800AD→900AD	+6900	+6850	Agrarian Consolidation	21,40M	+0,051%	260,7k km^2	12,183 m^2	14,90%	40,42%	34,40%
1000 AD	900AD→1000AD	+7000	+6950	Agrarian Consolidation	25,31M	+0,168%	301,9k km^2	11,929 m^2	14,90%	40,24%	34,25%

Table 5. (Continued) Regional paleodemographic trajectories across 21 historical epochs (10000 BC–1000 AD) for all five agricultural origins, packed across multi-panel pages.

Epoch	Interval	Point Δt	Mid Δt_{mid}	NDT Stage / Phase	Population (P)	Growth (r/yr)	Cropland Area	Cropland / Cap.	$p(0{-}4)$	Bocquet IJP $p(5{-}19)$	Immature / Total
Panel C: East Asia (Onset: 6000 BC (Yellow / Yangtze))											
10000 BC	—	-4000	—	Pre-NDT Foraging	329,12k	—	0 km^2	0 m^2	26,25%	64,60%	47,64%
9000 BC	10000BC→9000BC	-3000	-3500	Pre-NDT Foraging	519,99k	+0,046%	0 km^2	0 m^2	22,48%	61,25%	47,48%
8000 BC	9000BC→8000BC	-2000	-2500	Pre-NDT Foraging	726,02k	+0,033%	0 km^2	0 m^2	18,91%	48,09%	39,00%
7000 BC	8000BC→7000BC	-1000	-1500	Pre-NDT Foraging	1,70M	+0,085%	0 km^2	0 m^2	15,50%	37,22%	31,45%
6000 BC	7000BC→6000BC	+0	-500	Stage 1: NDT Boom	2,10M	+0,021%	1,3k km^2	602 m^2	18,93%	47,29%	38,34%
5000 BC	6000BC→5000BC	+1000	+500	Stage 2: Post-NDT Dispersal	1,64M	-0,025%	3,8k km^2	2,339 m^2	18,65%	47,64%	38,76%
4000 BC	5000BC→4000BC	+2000	+1500	Stage 2: Post-NDT Dispersal	2,59M	+0,046%	17,2k km^2	6,638 m^2	22,65%	61,03%	47,20%
3000 BC	4000BC→3000BC	+3000	+2500	Agrarian Consolidation	4,73M	+0,060%	40,8k km^2	8,627 m^2	25,44%	68,56%	51,12%
2000 BC	3000BC→2000BC	+4000	+3500	Agrarian Consolidation	6,42M	+0,031%	84,0k km^2	13,077 m^2	28,82%	84,83%	60,38%
1000 BC	2000BC→1000BC	+5000	+4500	Agrarian Consolidation	7,98M	+0,022%	93,6k km^2	11,730 m^2	51,30%	233,75%	113,84%
0 AD	1000BC→0AD	+6000	+5500	Agrarian Consolidation	64,63M	+0,209%	399,1k km^2	6,176 m^2	15,60%	43,42%	36,65%
100 AD	0AD→100AD	+6100	+6050	Agrarian Consolidation	58,20M	-0,105%	348,5k km^2	5,988 m^2	14,78%	39,74%	33,87%
200 AD	100AD→200AD	+6200	+6150	Agrarian Consolidation	65,65M	+0,120%	288,6k km^2	4,397 m^2	15,22%	42,59%	36,11%
300 AD	200AD→300AD	+6300	+6250	Agrarian Consolidation	58,06M	-0,123%	148,2k km^2	2,552 m^2	14,67%	38,73%	33,05%
400 AD	300AD→400AD	+6400	+6350	Agrarian Consolidation	52,84M	-0,094%	197,8k km^2	3,744 m^2	15,08%	41,31%	35,08%
500 AD	400AD→500AD	+6500	+6450	Agrarian Consolidation	53,81M	+0,018%	241,1k km^2	4,480 m^2	14,80%	40,31%	34,34%
600 AD	500AD→600AD	+6600	+6550	Agrarian Consolidation	52,64M	-0,022%	317,4k km^2	6,030 m^2	15,37%	43,69%	36,97%
700 AD	600AD→700AD	+6700	+6650	Agrarian Consolidation	52,82M	+0,003%	264,7k km^2	5,011 m^2	14,65%	39,89%	34,05%
800 AD	700AD→800AD	+6800	+6750	Agrarian Consolidation	55,51M	+0,050%	212,0k km^2	3,818 m^2	14,89%	41,73%	35,52%
900 AD	800AD→900AD	+6900	+6850	Agrarian Consolidation	55,07M	-0,008%	250,4k km^2	4,547 m^2	14,71%	40,37%	34,43%
1000 AD	900AD→1000AD	+7000	+6950	Agrarian Consolidation	67,16M	+0,198%	230,8k km^2	3,436 m^2	14,87%	40,57%	34,53%
Panel D: Southeast Asia (Onset: 2500 BC (Austroasiatic / Austronesian))											
10000 BC	—	-7500	—	Pre-NDT Foraging	141,68k	—	0 km^2	0 m^2	12,07%	25,69%	22,59%
9000 BC	10000BC→9000BC	-6500	-7000	Pre-NDT Foraging	172,83k	+0,020%	0 km^2	0 m^2	11,99%	29,02%	25,54%
8000 BC	9000BC→8000BC	-5500	-6000	Pre-NDT Foraging	186,39k	+0,008%	0 km^2	0 m^2	12,03%	28,18%	24,79%
7000 BC	8000BC→7000BC	-4500	-5000	Pre-NDT Foraging	337,78k	+0,059%	0 km^2	0 m^2	9,98%	22,27%	20,05%
6000 BC	7000BC→6000BC	-3500	-4000	Pre-NDT Foraging	324,29k	-0,004%	0,2 km^2	1 m^2	8,22%	17,94%	16,47%
5000 BC	6000BC→5000BC	-2500	-3000	Pre-NDT Foraging	197,56k	-0,050%	0,7 km^2	4 m^2	6,73%	14,89%	13,88%
4000 BC	5000BC→4000BC	-1500	-2000	Pre-NDT Foraging	264,17k	+0,029%	3,5 km^2	13 m^2	5,85%	12,68%	11,94%
3000 BC	4000BC→3000BC	-500	-1000	Pre-NDT Foraging	394,86k	+0,040%	8,0 km^2	20 m^2	10,92%	23,02%	20,51%
2000 BC	3000BC→2000BC	+500	+0	Stage 1: NDT Boom	701,32k	+0,057%	481,3 km^2	686 m^2	11,07%	24,38%	21,68%
1000 BC	2000BC→1000BC	+1500	+1000	Stage 2: Post-NDT Dispersal	1,63M	+0,084%	1,9k km^2	1,181 m^2	12,88%	30,44%	26,52%
0 AD	1000BC→0AD	+2500	+2000	Agrarian Consolidation	5,50M	+0,122%	6,1k km^2	1,110 m^2	14,87%	36,92%	31,44%
100 AD	0AD→100AD	+2600	+2550	Agrarian Consolidation	4,62M	-0,175%	10,9k km^2	2,359 m^2	14,42%	35,48%	30,36%
200 AD	100AD→200AD	+2700	+2650	Agrarian Consolidation	5,15M	+0,109%	12,0k km^2	2,336 m^2	14,78%	37,79%	32,20%
300 AD	200AD→300AD	+2800	+2750	Agrarian Consolidation	5,02M	-0,025%	13,3k km^2	2,639 m^2	14,45%	35,75%	30,58%
400 AD	300AD→400AD	+2900	+2850	Agrarian Consolidation	5,28M	+0,050%	14,6k km^2	2,770 m^2	14,88%	38,94%	33,15%
500 AD	400AD→500AD	+3000	+2950	Agrarian Consolidation	6,00M	+0,127%	16,1k km^2	2,686 m^2	14,71%	38,00%	32,41%
600 AD	500AD→600AD	+3100	+3050	Agrarian Consolidation	6,15M	+0,025%	17,8k km^2	2,887 m^2	15,18%	40,90%	34,69%
700 AD	600AD→700AD	+3200	+3150	Agrarian Consolidation	6,13M	-0,003%	19,6k km^2	3,191 m^2	14,55%	37,47%	32,01%
800 AD	700AD→800AD	+3300	+3250	Agrarian Consolidation	6,60M	+0,073%	21,6k km^2	3,273 m^2	14,89%	39,61%	33,71%
900 AD	800AD→900AD	+3400	+3350	Agrarian Consolidation	7,06M	+0,068%	23,6k km^2	3,344 m^2	14,70%	38,33%	32,70%
1000 AD	900AD→1000AD	+3500	+3450	Agrarian Consolidation	8,13M	+0,141%	25,8k km^2	3,176 m^2	14,53%	38,06%	32,53%

Table 5. (Continued) Regional paleodemographic trajectories across 21 historical epochs (10000 BC–1000 AD) for all five agricultural origins, packed across multi-panel pages.

Epoch	Interval	Point Δt	Mid Δt_{mid}	NDT Stage / Phase	Population (P)	Growth (r/yr)	Cropland Area	Cropland / Cap.	$p(0–4)$	Bocquet IJP $p(5–19)$	Immature / Total
Panel E: American Southwest (Onset: 0 AD (Basketmaker II/III Sedentary Village))											
10000 BC	—	-10000	—	Pre-NDT Foraging	19,11k	—	0 km²	0 m²	5,68%	10,75%	10,14%
9000 BC	10000BC→9000BC	-9000	-9500	Pre-NDT Foraging	21,42k	+0,011%	0 km²	0 m²	6,36%	14,12%	13,22%
8000 BC	9000BC→8000BC	-8000	-8500	Pre-NDT Foraging	24,13k	+0,012%	0 km²	0 m²	6,38%	13,54%	12,68%
7000 BC	8000BC→7000BC	-7000	-7500	Pre-NDT Foraging	44,52k	+0,061%	0 km²	0 m²	8,73%	18,88%	17,23%
6000 BC	7000BC→6000BC	-6000	-6500	Pre-NDT Foraging	47,11k	+0,006%	0 km²	0 m²	7,76%	16,91%	15,60%
5000 BC	6000BC→5000BC	-5000	-5500	Pre-NDT Foraging	31,76k	-0,039%	0 km²	0 m²	5,22%	11,49%	10,89%
4000 BC	5000BC→4000BC	-4000	-4500	Pre-NDT Foraging	42,60k	+0,029%	0 km²	0 m²	4,64%	10,08%	9,61%
3000 BC	4000BC→3000BC	-3000	-3500	Pre-NDT Foraging	103,81k	+0,089%	0 km²	0 m²	5,59%	11,58%	10,94%
2000 BC	3000BC→2000BC	-2000	-2500	Pre-NDT Foraging	115,11k	+0,010%	0 km²	0 m²	5,54%	11,58%	10,94%
1000 BC	2000BC→1000BC	-1000	-1500	Pre-NDT Foraging	174,13k	+0,041%	0 km²	0 m²	4,20%	8,80%	8,43%
0 AD	1000BC→0AD	+0	-500	Stage 1: NDT Boom	427,78k	+0,090%	175,4 km²	410 m²	14,74%	36,27%	30,92%
100 AD	0AD→100AD	+100	+50	Stage 1: NDT Boom	396,22k	-0,077%	183,8 km²	464 m²	14,27%	34,83%	29,86%
200 AD	100AD→200AD	+200	+150	Stage 1: NDT Boom	456,58k	+0,142%	192,6 km²	422 m²	14,59%	36,80%	31,43%
300 AD	200AD→300AD	+300	+250	Stage 1: NDT Boom	443,89k	-0,028%	201,8 km²	455 m²	14,23%	34,57%	29,65%
400 AD	300AD→400AD	+400	+350	Stage 1: NDT Boom	451,57k	+0,017%	211,5 km²	468 m²	14,69%	37,39%	31,89%
500 AD	400AD→500AD	+500	+450	Stage 1: NDT Boom	486,44k	+0,074%	221,6 km²	456 m²	14,45%	36,00%	30,80%
600 AD	500AD→600AD	+600	+550	Stage 1: NDT Boom	515,87k	+0,059%	232,2 km²	450 m²	14,97%	39,06%	33,21%
700 AD	600AD→700AD	+700	+650	Stage 1: NDT Boom	520,05k	+0,008%	243,3 km²	468 m²	14,29%	35,02%	30,02%
800 AD	700AD→800AD	+800	+750	Stage 1: NDT Boom	543,76k	+0,045%	255,2 km²	469 m²	14,59%	36,94%	31,55%
900 AD	800AD→900AD	+900	+850	Stage 1: NDT Boom	560,91k	+0,031%	287,5 km²	513 m²	14,39%	35,59%	30,47%
1000 AD	900AD→1000AD	+1000	+950	Stage 2: Post-NDT Dispersal	583,08k	+0,039%	320,0 km²	549 m²	14,31%	35,10%	30,08%

Note: Population and cropland areas use semantic abbreviations (k = thousands, M = millions) and European decimals.

References

[1] Klein Goldewijk, Kees, et al. "Anthropogenic Land Use Estimates for the Holocene – HYDE 3.2." *Earth System Science Data*, vol. 9, no. 2, 2017, pp. 927–53, https://doi.org/10.5194/essd-9-927-2017.

[2] Ayer, Miriam, et al. "An Empirical Distribution Function for Sampling with Incomplete Information." *The Annals of Mathematical Statistics*, vol. 26, no. 4, 1955, pp. 641–47, https://doi.org/10.1214/aoms/1177728423.

[3] Whittaker, Edmund T. "On a New Method of Graduation." *Proceedings of the Edinburgh Mathematical Society*, vol. 41, 1923, pp. 63–75, https://doi.org/10.1017/S0013091500077853.

[4] Human Mortality Database. "Human Mortality Database: University of California, Berkeley, and Max Planck Institute for Demographic Research." Max Planck Institute for Demographic Research, 2024, https://www.mortality.org.

[5] United Nations, Department of Economic and Social Affairs, Population Division. *World Population Prospects 2024: Methodology of the United Nations Population Estimates and Projections*. United Nations, 2024, https://population.un.org/wpp/.

[6] Niva, Venla, et al. "World's Human Migration Patterns in 2000–2019 Unveiled by High-Resolution Data." *Nature Human Behaviour*, vol. 7, no. 11, 2023, pp. 2023–37, https://doi.org/10.1038/s41562-023-01689-4.

[7] Bocquet-Appel, Jean-Pierre. "Paleoanthropological Traces of a Neolithic Demographic Transition." *Current Anthropology*, vol. 43, no. 4, 2002, pp. 637–50, https://doi.org/10.1086/342428.

[8] Bocquet-Appel, Jean-Pierre. "Explaining the Neolithic Demographic Transition." *The Neolithic Demographic Transition and Its Consequences*, edited by Jean-Pierre Bocquet-Appel and Ofer Bar-Yosef, Springer, 2008, pp. 35–55, https://doi.org/10.1007/978-1-4020-8539-0_3.

[9] Shennan, Stephen, et al. "Regional Population Collapse Followed Initial Agriculture Booms in Mid-Holocene Europe." *Nature Communications*, vol. 4, no. 1, 2013, p. 2486, https://doi.org/10.1038/ncomms3486.

[10] Timpson, Adrian, et al. "Reconstructing Regional Population Fluctuations in the European Neolithic Using Radiocarbon Dates: A New Case-Study Using an Improved Method." *Journal of Archaeological Science*, vol. 52, 2014, pp. 549–57, https://doi.org/10.1016/j.jas.2014.08.011.